

Olympiad Test Paper — Mathematics (Class 7)

Chapter 5: Parallel and Intersecting Lines

Subject	Mathematics (NCERT Class 7)
Chapter	5 — Parallel and Intersecting Lines
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Read each diagram description carefully and label the angles yourself. Remember the three big traps: corresponding/alternate angles are equal ONLY when the lines are parallel; co-interior (same-side interior) angles are SUPPLEMENTARY (add to 180°), not equal; and only PERPENDICULAR crossings give four equal angles. Chase angles step by step using linear pairs and vertically-opposite equalities — you rarely need a protractor. Do not write on this paper.

1. Two distinct straight lines in a plane can intersect in at most how many points?

- (A) 1 (B) 0
(C) 2 (D) infinitely many

2. Two straight lines cross at a point, forming four angles. One of them is 110° . The angle vertically opposite to it measures:

- (A) 70° (B) 90°
(C) 110° (D) 250°

3. Two lines intersect and one of the four angles is 110° . The angle adjacent to it (its linear pair) is:

- (A) 110° (B) 90°
(C) 80° (D) 70°

4. When two lines cross so that all four angles are equal, each angle is $360 \div 4$. Such lines are called:

- (A) parallel (B) perpendicular
(C) transversal (D) co-interior

5. Two adjacent angles on a straight line form a linear pair. If one is 4 times the other, the larger angle is:

- (A) 36° (B) 120°
(C) 144° (D) 150°

6. Opposite edges of a rectangular sheet of paper are _____ and adjacent edges are _____.

- (A) perpendicular; parallel (B) parallel; perpendicular
(C) parallel; parallel (D) perpendicular; perpendicular

7. A transversal crosses two lines at two distinct points. The total number of angles formed is:

- (A) 4 (B) 6
(C) 8 (D) 12

8. A transversal cuts two parallel lines. A corresponding angle at the first line is 73° . The matching corresponding angle at the second line is:

- (A) 17° (B) 107°
(C) 287° (D) 73°

9. A transversal cuts two parallel lines. One co-interior (same-side interior) angle is 118° . The other co-interior angle is:

- (A) 118° (B) 62°
(C) 72° (D) 242°

10. Two lines cross. One angle is 35° . Its vertically opposite angle and its linear-pair neighbour are, respectively:

- (A) 35° and 145° (B) 35° and 35°
(C) 145° and 145° (D) 55° and 145°

11. Two intersecting lines form four angles a, b, c, d in order around the point. Which statement is ALWAYS true, whatever the lines?

- (A) $a = b$ (B) $a = c$
(C) $a = b = c = d$ (D) $a + c = 180^\circ$

12. A transversal cuts two parallel lines, making just two distinct angle measures. If one measure is 72° , the two measures are:

- (A) 72° and 108° (B) 72° and 90°
(C) 72° and 18° (D) 72° only

13. Line $a \perp$ line b , and line $b \parallel$ line c . The angle between line a and line c is:

- (A) 0° (B) 45°
(C) 90° (D) 180°

14. At an intersection of two lines, one angle is x° and an adjacent angle is $(x + 40)^\circ$. The value of x is:

- (A) 80 (B) 110
(C) 90 (D) 70

15. Two parallel lines are cut by a transversal. A pair of alternate interior angles measure $(2x + 10)^\circ$ and $(3x - 20)^\circ$. The value of x is:

- (A) 30 (B) 10
(C) 6 (D) 38

16. A transversal cuts two parallel lines p and q . An angle of 65° is formed at p . Its corresponding angle, an alternate angle, and a co-interior angle to it measure, respectively:

- (A) 65° , 115° , 115° (B) 115° , 65° , 65°
(C) 65° , 65° , 115° (D) 65° , 65° , 65°

17. Which of these conditions guarantees that two lines cut by a transversal are parallel?

- (A) a pair of co-interior angles add to 200° (B) a pair of alternate angles are equal
(C) a pair of co-interior angles are equal
(D) the transversal is perpendicular to only one of the lines

18. Two parallel lines are cut by a transversal. One co-interior angle is twice the other. The smaller co-interior angle is:

- (A) 30° (B) 90°
(C) 120° (D) 60°

19. Two lines intersect. The sum of the two angles that are vertically opposite to each other in ONE such pair (i.e. an angle plus its vertically-opposite partner) is 86° . Each of these two angles measures:

- (A) 43° (B) 86°
(C) 94° (D) 172°

20. Two parallel lines are cut by a transversal. Two corresponding angles are $(5x)^\circ$ and $(3x + 40)^\circ$. The common value of each corresponding angle is:

- (A) 20° (B) 100°
(C) 60° (D) 140°

21. Which statement is FALSE?

- (A) Vertically opposite angles are always equal.
(B) Corresponding angles are equal only when the two lines are parallel.
(C) Co-interior angles between two parallel lines are equal.
(D) Adjacent angles on a straight line add to 180° .

22. A transversal cuts two lines (NOT stated to be parallel). At the first line a pair of corresponding positions has 80° ; at the second line the corresponding angle is measured to be 95° . From this you can conclude the two lines are:

- (A) parallel (B) perpendicular
(C) coincident (D) not parallel

23. Three angles around the meeting point of two intersecting lines are marked $2y$, $3y$, and $2y$ reading around. The fourth angle equals:

- (A) $2y$ (B) $3y$
(C) $5y$ (D) y

24. In the figure, lines AB and CD cross at O. $\angle AOC = (3x - 10)^\circ$ and $\angle BOD = (2x + 25)^\circ$ are vertically opposite. Then $\angle AOC$ measures:

- (A) 35° (B) 60°
(C) 95° (D) 105°

25. A transversal t cuts parallel lines m and n . At m it makes an angle of 40° with the transversal on one side. The acute angle the transversal makes with n is:

- (A) 40° (B) 50°
(C) 140° (D) 20°

26. Two parallel lines are cut by a transversal. The two co-interior (same-side interior) angles measure $(4x)^\circ$ and $(5x - 18)^\circ$. Then x equals:

- (A) 18 (B) 22
(C) 20 (D) 30

27. Ray OP stands on a line, making two angles $3x$ and $2x$ on either side. Then x is:

- (A) 18° (B) 60°
(C) 72° (D) 36°

28. A transversal cuts two parallel lines. One of the eight angles is 124° . How many of the eight angles equal 124° ?

- (A) 1 (B) 2
(C) 4 (D) 8

29. Two lines are perpendicular. A third line is parallel to the first. The angle between this third line and the second line is:

- (A) 90° (B) 45°
(C) 0° (D) cannot be found

30. Lines l and m are parallel, cut by transversal t . $\angle 1$ (above l , right of t) = 118° . Its alternate interior angle (below l) pairs with the interior angle at m . The alternate interior angle measure is:

- (A) 62° (B) 118°
(C) 90° (D) 236°

31. At a crossing of two lines, the four angles in order are p , q , p , q . Given $p = 3q$, then q equals:

- (A) 30° (B) 60°
(C) 135° (D) 45°

32. A transversal makes co-interior angles $(x + 30)^\circ$ and $(2x)^\circ$ with two parallel lines. The larger co-interior angle is:

- (A) 50° (B) 80°
(C) 100° (D) 130°

33. Two angles are both supplementary AND equal to each other. Each angle is:

- (A) 90° (B) 60°
(C) 45° (D) 180°

34. A transversal cuts two parallel lines. The bisector of one of the co-interior angles and the bisector of the other co-interior angle meet at an angle of:

- (A) 45° (B) 60°
(C) 90° (D) 120°

35. In a figure, line $PQ \parallel$ line RS , and a transversal meets PQ at A and RS at B . $\angle PAB = 5x^\circ$ and $\angle ABS = 4x^\circ$ are co-interior angles. Then $\angle PAB$ measures:

- (A) 20° (B) 80°
(C) 120° (D) 100°

36. Which of the following pairs of angles need NOT be equal even when two lines are cut by a transversal?

- (A) vertically opposite angles
(B) corresponding angles when the lines are not parallel
(C) angles in a linear pair that are both right angles
(D) two angles each equal to 90°

37. Two parallel lines are cut by a transversal so that a pair of alternate angles are $(60 - x)^\circ$ and $(2x)^\circ$. Each of these alternate angles measures:

- (A) 40° (B) 20°
(C) 60° (D) 120°

38. Lines a and b are parallel. A transversal makes an interior angle of 105° with a on one side. The interior angle with b on the SAME side of the transversal is:

- (A) 165° (B) 105°
(C) 75° (D) 90°

39. At an intersection, an angle and its vertically opposite angle are $(4x + 5)^\circ$ and $(6x - 25)^\circ$. The value of x is:

- (A) 10 (B) 15
(C) 20 (D) 30

40. A transversal cuts lines u and v . Corresponding angles are 88° (at u) and 88° (at v). What can you conclude?

- (A) u and v are perpendicular (B) u and v meet at 88°
(C) nothing at all (D) u and v are parallel

41. Lines $l \parallel m$, cut by transversal t . One interior angle on the left of t at l is $7x^\circ$ and the co-interior angle at m is $2x^\circ$. Then the angle $7x^\circ$ measures:

- (A) 20° (B) 40°
(C) 140° (D) 120°

42. Two lines intersect, and one of the four angles is a right angle. The number of the four angles that are right angles is:

- (A) 4 (B) 2
(C) 3 (D) 1

43. A transversal cuts two parallel lines. An exterior angle on one line is 130° . The interior angle on the same side of the transversal, at that SAME crossing (its linear pair), is:

- (A) 130° (B) 50°
(C) 65° (D) 230°

44. Parallel lines l and m are cut by transversal t . $\angle a$ and $\angle b$ are alternate exterior angles, with $\angle a = (3x + 15)^\circ$ and $\angle b = (4x - 5)^\circ$. Then $\angle a$ measures:

- (A) 20° (B) 60°
(C) 105° (D) 75°

45. Two roads cross. The angle in the north-east "corner" between them is 65° . The angle in the south-west corner is:

- (A) 25° (B) 115°
(C) 65° (D) 130°

46. A transversal cuts two parallel lines, and one of the eight angles is a right angle. Then:

- (A) all eight angles are right angles (B) exactly two are right angles
(C) exactly four of the eight angles are right angles
(D) only that one angle is a right angle

47. Two parallel lines are cut by a transversal. The difference between a pair of co-interior angles is 40° . The larger co-interior angle is:

- (A) 70° (B) 110°
(C) 90° (D) 140°

48. Lines AB and CD intersect at O . $\angle AOC = 50^\circ$. Ray OE is drawn so that $\angle COE = 90^\circ$ (OE between OC and OB region). Then $\angle EOB$ equals:

- (A) 130° (B) 50°
(C) 90° (D) 40°

49. A transversal cuts parallel lines so that an alternate interior angle is 3 times its co-interior partner on the same line. (The alternate interior angle and the co-interior angle here form a linear pair on the cut line.) The alternate interior angle is:

- (A) 135° (B) 90°
(C) 120° (D) 45°

50. Which set of three angle measures could be the three angles among the four formed by two intersecting lines (the fourth being the remaining one)?

(A) 60° , 60° , 60°

(B) 100° , 100° , 100°

(C) 50° , 130° , 50°

(D) 40° , 40° , 140°

51. Lines $p \parallel q$. A transversal makes a 3:2 ratio between a pair of co-interior angles. The smaller of the two angles is:

(A) 36°

(B) 72°

(C) 54°

(D) 108°

52. A transversal crosses two parallel lines at exactly 90° . The number of distinct angle measures among the eight angles is:

(A) 8

(B) 2

(C) 4

(D) 1

53. $\angle 1$ and $\angle 2$ are corresponding angles for two parallel lines, and $\angle 2$ and $\angle 3$ form a linear pair. If $\angle 1 = 116^\circ$, then $\angle 3$ equals:

(A) 64°

(B) 116°

(C) 90°

(D) 26°

54. Two parallel lines are cut by a transversal. An alternate-interior angle and a co-interior angle that share the same vertex on one of the lines satisfy: their measures are in ratio 2:1. Each unknown — the alternate-interior angle — measures:

(A) 60°

(B) 90°

(C) 120°

(D) 135°

55. Lines m and n are crossed by a transversal. Corresponding angles measure $(x + 25)^\circ$ at m and $(2x - 15)^\circ$ at n , and $m \parallel n$. Then the corresponding angle measure is:

(A) 40°

(B) 65°

(C) 50°

(D) 80°

56. A transversal cuts two parallel lines l and m . $\angle A$ on l and $\angle B$ on m are a pair of alternate angles, and $\angle A$ is 30° less than twice $\angle B$. Then $\angle A$ measures:

(A) 15°

(B) 45°

(C) 60°

(D) 30°

57. Two intersecting lines make four angles; three of them are $(x)^\circ$, $(x)^\circ$, and $(2x)^\circ$. The value of x is:

(A) 60

(B) 45

(C) 72

(D) 90

58. Lines $a \parallel b$, cut by a transversal. The bisector of one of the corresponding angles and the line b enclose what angle, if the corresponding angle is 80° ? (The bisector makes 40° with the transversal; find the acute angle it makes with b .)

(A) 20°

(B) 80°

(C) 40°

(D) 60°

59. A transversal cuts two lines. If a pair of co-interior angles measure 100° and 90° , the two lines are:

(A) parallel

(B) not parallel

(C) perpendicular

(D) coincident

60. Parallel lines l and m are cut by a transversal. $\angle P$ is an interior angle at l and $\angle Q$ is the alternate interior angle at m . A second transversal cuts the same lines making $\angle Q$'s alternate angle equal to $\angle P$. If $\angle P = 67^\circ$, then the alternate interior angle $\angle Q$ equals:

(A) 23°

(B) 113°

(C) 134°

(D) 67°

End of question paper. Maximum marks: 240. Marking: +4 correct, -1 wrong, 0 unattempted.

Solutions — Mathematics (Class 7), Chapter 5: Parallel and Intersecting Lines

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	B	21	C	31	D	41	C	51	B
2	C	12	A	22	D	32	C	42	A	52	D
3	D	13	C	23	B	33	A	43	B	53	A
4	B	14	D	24	C	34	C	44	D	54	C
5	C	15	A	25	A	35	D	45	C	55	B
6	B	16	C	26	B	36	B	46	A	56	D
7	C	17	B	27	D	37	A	47	B	57	A
8	D	18	D	28	C	38	C	48	D	58	C
9	B	19	A	29	A	39	B	49	A	59	B
10	A	20	B	30	B	40	D	50	C	60	D

Answer-key distribution: A = 14, B = 16, C = 16, D = 14.

Worked Solutions

1. (A) Two distinct straight lines, once they touch, immediately separate (a straight line can't bend back to meet again). So they share **at most one** point. (Zero if parallel, one if intersecting — but "at most" is 1.)

2. (C) Vertically opposite angles are always equal, regardless of how the lines cross. So the angle opposite 110° is **110°** . Trap: 70° is the *adjacent* (linear-pair) angle, not the opposite.

3. (D) Adjacent angles on a straight line form a linear pair summing to 180° . So $180^\circ - 110^\circ = \mathbf{70^\circ}$. Trap (A) 110° confuses adjacent with vertically opposite.

4. (B) All four equal $\Rightarrow$ each $= 360 \div 4 = 90^\circ$. Lines crossing at right angles are **perpendicular**.

5. (C) Let the smaller be x ; the larger is $4x$. Linear pair: $x + 4x = 180 \Rightarrow 5x = 180 \Rightarrow x = 36$. Larger $= 4 \cdot 36 = \mathbf{144^\circ}$. Trap (A) 36° is the *smaller* angle.

6. (B) Opposite edges of a sheet run the same way and never meet — **parallel**. Adjacent edges meet at a 90° corner — **perpendicular**.

7. (C) Each crossing of the transversal makes 4 angles; two crossings $\Rightarrow 4 \times 2 = 8$ angles.

8. (D) Corresponding angles are equal when the lines are parallel, so the matching angle is 73° . Traps: 107° is its co-interior/linear-pair partner; 17° is its complement (irrelevant here).

9. (B) Co-interior (same-side interior) angles are supplementary on parallel lines: $180^\circ - 118^\circ = 62^\circ$. Trap (A) 118° wrongly assumes they are equal.

10. (A) Vertically opposite to 35° is 35° ; the linear-pair neighbour is $180^\circ - 35^\circ = 145^\circ$. So **35° and 145°** .

11. (B) Whatever the lines, opposite angles are equal: $a = c$ (and $b = d$). $a = b$ only if perpendicular; $a + c = 180^\circ$ is false in general ($a + b = 180^\circ$ is the true relation). So **$a = c$** .

12. (A) A transversal on parallel lines gives angles of just two sizes that are supplementary: 72° and $180^\circ - 72^\circ = 108^\circ$. So **72° and 108°** . Trap (B) $72^\circ + 90^\circ$ and (C) $72^\circ + 18^\circ$ misuse complement/right-angle ideas.

13. (C) $a \perp b$ means 90° between them. $b \parallel c$ means c has b 's direction, so a meets c at the same 90° . Answer **90°** .

14. (D) Adjacent angles on a line: $x + (x + 40) = 180 \Rightarrow 2x = 140 \Rightarrow x = 70$. Trap (C) 110 is the value of the second angle, not x .

15. (A) Alternate interior angles are equal: $2x + 10 = 3x - 20 \Rightarrow 30 = x \Rightarrow x = 30$. (Then each angle = 70° .) Trap (B) 10 comes from wrongly setting the sum to 180.

16. (C) Corresponding = 65° (equal), alternate = 65° (equal), co-interior = $180^\circ - 65^\circ = 115^\circ$ (supplementary). So **65° , 65° , 115°** . Trap (D) wrongly makes the co-interior angle equal too.

17. (B) Equal alternate angles force the lines parallel. (A) co-interior must sum to $*180^\circ*$, not 200° ; (C) equal co-interior would mean each 90° only — not a general parallel test; (D) is about one line only. Answer **alternate angles equal**.

18. (D) Co-interior angles are supplementary. Let the smaller be x , larger $2x$: $x + 2x = 180 \Rightarrow x = 60$. Smaller = **60°** . Trap (A) 30 ignores that they must add to 180° .

19. (A) An angle and its vertically opposite partner are equal, so each = $86^\circ \div 2 = 43^\circ$. Trap (B) 86° forgets to halve.

20. (B) Corresponding angles equal: $5x = 3x + 40 \Rightarrow 2x = 40 \Rightarrow x = 20$, so each = $5 \cdot 20 = 100^\circ$. Trap (A) 20° gives x , not the angle.

21. (C) Co-interior angles between parallel lines are **supplementary, not equal** — so "they are equal" is the FALSE statement. The other three are all true.

22. (D) Corresponding angles are equal *only* when the lines are parallel. Here they differ (80° vs 95°), so the lines are **not parallel**. Trap (A) reverses the rule.

23. (B) Going round, the 1st and 3rd (both $2y$) are a vertically opposite pair (equal ✓). The remaining pair is the 2nd and 4th, which are also vertically opposite, so the 4th = the 2nd = $3y$.

24. (C) Vertically opposite $\Rightarrow 3x - 10 = 2x + 25 \Rightarrow x = 35$. $\angle AOC = 3 \cdot 35 - 10 = 95^\circ$. Trap (A) 35 reports x instead of the angle.

25. (A) The transversal cuts the parallels at the same inclination, so the acute angle at n equals the acute angle at $m = 40^\circ$ (corresponding/alternate equality). Trap (B) 50° is the complement.

26. (B) Co-interior angles are supplementary: $4x + (5x - 18) = 180 \Rightarrow 9x = 198 \Rightarrow x = 22$. Trap (B-vs-C): setting them *equal* gives $x = 18$ (option A), the classic co-interior slip.

27. (D) Ray OP on a line $\Rightarrow$ the two angles form a linear pair: $3x + 2x = 180 \Rightarrow 5x = 180 \Rightarrow x = 36^\circ$. Trap (B) 60° comes from dividing 180 by 3.

28. (C) Eight angles take two supplementary sizes, four of each. So **4** of them equal 124° (its vertically opposite + the two corresponding-position equals). Trap (B) 2 forgets the corresponding pair at the other line.

29. (A) Perpendicular lines: 90° between first and second. A line parallel to the first keeps its direction, so it meets the second at the same 90° .

30. (B) Alternate interior angles are equal on parallel lines, so the alternate of 118° is 118° . Trap (A) 62° wrongly treats them as supplementary (that's co-interior).

31. (D) Around the point the four angles are p, q, p, q with $p + q = 180^\circ$ (adjacent linear pair). With $p = 3q$: $3q + q = 180 \Rightarrow 4q = 180 \Rightarrow q = 45^\circ$.

32. (C) Co-interior supplementary: $(x + 30) + 2x = 180 \Rightarrow 3x = 150 \Rightarrow x = 50$. The two angles are 80° and 100° ; the larger is **100°** . Trap (B) 80° is the smaller.

33. (A) "Both supplementary AND equal": $x + x = 180 \Rightarrow x = 90^\circ$. Trap (D) 180° is the sum, not each angle.

34. (C) The two co-interior angles sum to 180° ; their halves sum to 90° . In the triangle formed by the two bisectors and the transversal, the bisectors meet at $180^\circ - 90^\circ = \mathbf{90^\circ}$. (A neat classic result.)

35. (D) $\angle PAB$ and $\angle ABS$ are co-interior $\Rightarrow$ supplementary: $5x + 4x = 180 \Rightarrow 9x = 180 \Rightarrow x = 20$. $\angle PAB = 5 \cdot 20 = \mathbf{100^\circ}$. Trap (B) 80° is $\angle ABS$.

36. (B) Vertically opposite angles, right-angle linear pairs, and two 90° angles are always equal. But **corresponding angles when the lines are NOT parallel** need not be equal — that's the answer.

37. (A) Alternate angles equal: $60 - x = 2x \Rightarrow 60 = 3x \Rightarrow x = 20$, so each = $2 \cdot 20 = \mathbf{40^\circ}$. Trap (B) 20° reports x , not the angle.

38. (C) The SAME-side interior angle pairs as co-interior with 105° , so it is supplementary: $180^\circ - 105^\circ = \mathbf{75^\circ}$. Trap (B) 105° wrongly treats them as equal (that would be alternate, opposite side).

39. (B) Vertically opposite $\Rightarrow$ equal: $4x + 5 = 6x - 25 \Rightarrow 30 = 2x \Rightarrow x = \mathbf{15}$. Trap (A) 10 comes from a sign slip.

40. (D) Equal corresponding angles guarantee the lines are **parallel** (the converse rule). Trap (C) "meet at 88° " misreads the equality.

41. (C) The given interior angle at l and the co-interior angle at m are supplementary: $7x + 2x = 180 \Rightarrow 9x = 180 \Rightarrow x = 20$. So $7x = \mathbf{140^\circ}$. Trap (D) $120^\circ = 6x$, an arithmetic slip.

42. (A) If one of the four angles is 90° , its linear-pair neighbour is 90° , its vertically opposite is 90° , and the fourth is 90° too. So **4** of them are right angles.

43. (B) The exterior angle (130°) and the interior angle at the same crossing are a linear pair: $180^\circ - 130^\circ = \mathbf{50^\circ}$. Trap (A) 130° forgets the straight-line sum.

44. (D) Alternate exterior angles are equal on parallel lines: $3x + 15 = 4x - 5 \Rightarrow 20 = x$. $\angle a = 3 \cdot 20 + 15 = \mathbf{75^\circ}$. Trap (A) 20 reports x .

45. (C) The NE corner and the SW corner are vertically opposite, hence equal: $\mathbf{65^\circ}$. Trap (B/D) $115^\circ/130^\circ$ treat them as supplementary/double.

46. (A) With a right angle present, that whole crossing becomes four 90° angles, and corresponding angles at the parallel line are 90° too — so **all eight** angles are right angles. (This is why a perpendicular transversal makes a fully right-angled grid.)

47. (B) Co-interior sum 180° , difference 40° . Larger = $(180 + 40) \div 2 = \mathbf{110^\circ}$ (smaller 70°). Trap (A) 70° is the smaller angle.

48. (D) A, O, B are collinear so $\angle AOB = 180^\circ$. $\angle AOC + \angle COE + \angle EOB = 180^\circ \Rightarrow 50^\circ + 90^\circ + \angle EOB = 180^\circ \Rightarrow \angle EOB = 40^\circ$. Trap (B) 50° just echoes $\angle AOC$.

49. (A) The alternate-interior angle and the co-interior angle sharing that vertex form a linear pair (sum 180°). With alternate = $3 \times$ co-interior: $3k + k = 180 \Rightarrow k = 45 \Rightarrow$ alternate = 135° . Trap (D) 45° is the co-interior partner.

50. (C) Four angles around a crossing come in two equal vertically-opposite pairs, with adjacent ones supplementary. $50^\circ, 130^\circ, 50^\circ$ works: fourth angle = 130° , pairs $50/50$ and $130/130$, and $50 + 130 = 180$ ✓. (B) needs four equal 60° 's impossible; (D) 100° three times can't sum correctly.

51. (B) Co-interior angles sum to 180° in ratio $3:2 \Rightarrow 5$ parts = $180^\circ \Rightarrow 1$ part = 36° . Smaller = 2 parts = 72° . Trap (A) 36° is one part, not the smaller angle.

52. (D) At 90° every one of the eight angles is a right angle, so there is just **1** distinct measure (90°). Trap (B) 2 forgets the perpendicular case collapses both sizes to 90° .

53. (A) $\angle 1 = \angle 2 = 116^\circ$ (corresponding, parallel). $\angle 2$ and $\angle 3$ are a linear pair $\Rightarrow \angle 3 = 180^\circ - 116^\circ = 64^\circ$. Trap (B) 116° forgets the linear-pair step.

54. (C) The alternate-interior angle and the co-interior angle at that vertex form a linear pair (sum 180°) in ratio $2:1 \Rightarrow 2k + k = 180 \Rightarrow k = 60$. Alternate-interior = $2k = 120^\circ$. Trap (A) 60° is the co-interior partner.

55. (B) Corresponding angles equal ($m \parallel n$): $x + 25 = 2x - 15 \Rightarrow 40 = x$. Angle = $40 + 25 = 65^\circ$. Trap (A) 40 reports x .

56. (D) Alternate angles are equal: $\angle A = \angle B$. Given $\angle A = 2\angle B - 30 \Rightarrow \angle B = 2\angle B - 30 \Rightarrow \angle B = 30$, so $\angle A = 30^\circ$. Trap (C) 60° doubles incorrectly.

57. (A) Four angles around a crossing form two vertically-opposite pairs. Three given are $x, x, 2x$, so the pair x, x is one vertically-opposite pair and $2x$ belongs to the other (whose partner is also $2x$). Sum = $x + x + 2x + 2x = 6x = 360 \Rightarrow x = 60$. Check: $60^\circ + 60^\circ + 120^\circ + 120^\circ = 360^\circ$ ✓. Trap (D) 90 would force all four equal, contradicting $x \neq 2x$.

58. (C) The corresponding angle is 80° , so its bisector makes 40° with the transversal. Because the angle's other arm lies along line b 's direction... measuring the acute angle the bisector makes with b : $80^\circ - 40^\circ = 40^\circ$. Trap (D) 80° forgets the bisecting.

59. (B) For parallel lines co-interior angles must sum to 180° . Here $100^\circ + 90^\circ = 190^\circ \neq 180^\circ$, so the lines are **not parallel**. Trap (A) assumes parallel without checking the sum.

60. (D) $\angle Q$ is the alternate interior angle of $\angle P$ across the (first) transversal between the parallel lines, so $\angle Q = \angle P = 67^\circ$ (alternate interior angles equal). The second-transversal clause is consistent extra detail. Trap (C) 113° treats them as co-interior (supplementary).